
Liver-Muscle Axis in DLY vs TFB: Validated Signaling Model
Hepatokine → Receptor → Muscle Response | Full 9234-gene muscle validation | 7464 common genes
Legend
Liver DLY > TFB (positive FC)
Liver TFB > DLY (negative FC)
Receptor → Muscle target
Discordant/Complex
15 kg
Early Programming
45 kg
TFB Peak
75 kg
DLY Rising
105 kg
Late Divergence
Liver-Muscle r=0.38
Liver-Muscle r=0.20
Liver-Muscle r=0.16
Liver-Muscle r=0.08
LIVER
HEPATOKINES
IGF1
IGFBP1
IGFBP3
FGF21
FST
ANGPTL4
CCN2
APOE
MUSCLE
RESPONSE
r=0.95
r=-0.79
r=-0.04
r=0.05
r=0.92
r=-0.50
→ IGF1R
→ -
→ TMEM219
→ FGFR1/KLB
→ ACVR2B
→ ITGB1
→ ITGB1
→ LDLR
7464
genes shared
liver & muscle
0.38→0.08
cross-tissue r
(15kg→105kg)
21/37
receptor-ligand
pairs validated
IGF1 r=0.95
top coordinated
hepatokine
FST→ACVR2B
r=0.95
top receptor
correlation
4-STAGE MODEL: 15kg Genetic Programming → 45kg TFB Peak (compensatory FST/MSTN) → 75kg DLY Catch-up (AA loading divergence) → 105kg Accumulated Consequence
IGF1 Axis (validated): Liver DLY↑ → Muscle DLY↑ throughout 15-45kg | FST-MSTN Axis: TFB liver↑FST → ACVR2B↓ → MSTN signaling altered → muscle deposition ceiling
Key insight: liver-muscle transcriptional coupling STRONGEST at 15kg (r=0.38) and WEAKEST at 105kg (r=0.08) — programming sets trajectory, later stages reflect accumulated divergence
IGF1r=0.95: IGF1 is the strongest validated hepatokine.
Liver DLY↑ → Muscle DLY↑ synchronized at 15-45kg.
FST→ACVR2B r=0.95: Follistatin receptor tracks liver FST.
FST is a MSTN antagonist: TFB liver↑ → may suppress muscle growth via MSTN.
IGFBP1: Strongest liver signal (FC=-3.37, 15kg) but NOT in muscle.
Acts as serum IGF1 carrier → regulates IGF1 bioavailability systemically.
Cross-tissue r DECLINES: 0.38→0.08 over stages.
Liver-muscle coupling strongest early, decouples as deposition diverges.
ITGB1 is the most shared receptor (7 ligands).
Integrin β1 mediates ANGPTL4/6/8, CCN1/2, IGFBP1/2, SPARC signaling.